

BRAC University
Dept. of Computer Science and Engineering
Fall 2024
CSE331 - Assignment 1
Deadline: November 21 11:45 PM

1. Construct DFAs that recognize the following languages:
 - a. $L = \{ w \in \{0,1\}^* : w \text{ contains "1010" as a substring} \}$
 - b. $L = \{ w \in \{a,b\}^* : \text{"a" occurs in every 4th position} \}$
 - c. $L = \{ w \in \{a,b\}^* : \text{every "b" is followed by at most one "a"} \}$
 - d. $L = \{ w \in \{0,1\}^* : w \text{ starts and ends with 1 and length of } w \text{ is odd} \}$
 - e. $L = \{ w \in \{0,1\}^* : \text{the binary equivalent string is divisible by 3} \}$
2. Construct NFAs that recognize the following languages:
 - a. $L = \{ w \in \{0,1\}^* : w \text{ starts with "1010"} \}$
 - b. $L = \{ w \in \{0,1\}^* : \text{3rd last symbol in } w \text{ is 1} \}$
 - c. $L = \{ w \in \{a,b\}^* : w \text{ ends with "bab"} \}$
 - d. $L = \{ w \in \{a,b\}^* : \text{the count of substring "ab" in } w \text{ is at least two} \}$
 - e. $L = \{ w \in \{0,1\}^* : w \text{ contains "010" as a substring} \}$
3. Write regular expressions for the following languages:
 - a. $L = \{ w \in \{0,1\}^* : \text{length of } w \text{ is even} \}$
 - b. $L = \{ w \in \{a,b\}^* : w \text{ starts and ends with same symbol and the length of } w \text{ is odd} \}$
 - c. $L = \{ w \in \{0,1\}^* : w \text{ contains exactly one "01"} \}$
 - d. $L = \{ w \in \{a,b\}^* : \text{every "b" is followed by at least two "a"s} \}$
 - e. $L = \{ w \in \{0,1\}^* : w \text{ starts with "1011"} \}$

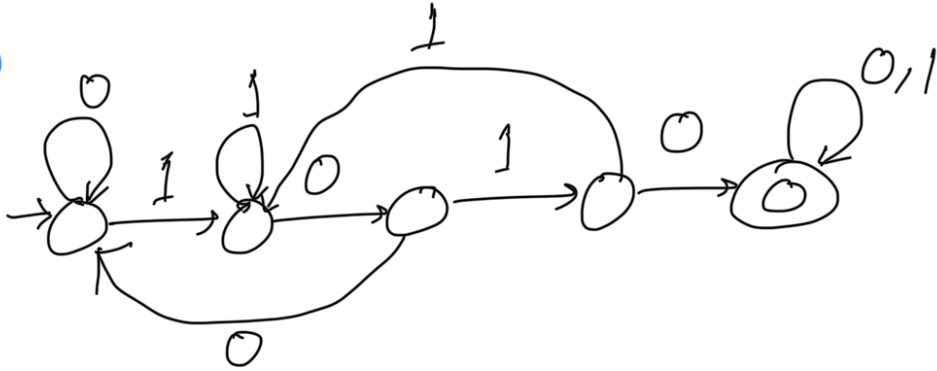
BRAC University
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Fall 2024
CSE331 - Assignment 2
Deadline: January 04 11:45 PM

1. Use the pumping lemma to demonstrate that the following languages are not regular.
- a. $L = \{ w \in \{0,1\}^* : ww^R \text{ and } w^R \text{ means } w \text{ in reverse} \}$
 - b. $L = \{ w \in \{a,b\}^* : w = b^n a^m \text{ where } n > m, m \geq 0 \}$
 - c. $L = \{ w \in \{0,1,2,3\}^* : w = 1^n 0^m 3^n 2^m \text{ where } n, m \geq 0 \}$
 - d. $L = \{ w \in \{0,1\}^* : w = 1^n : n \text{ is a power of three} \}$

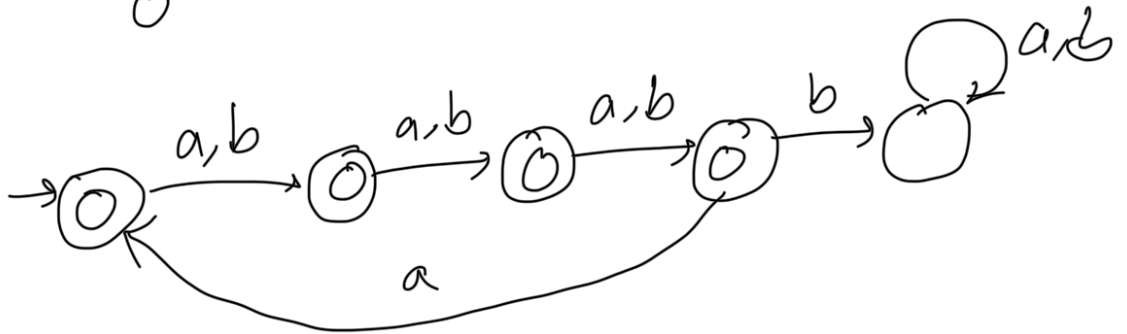
Assignment 1: Solutions

1.

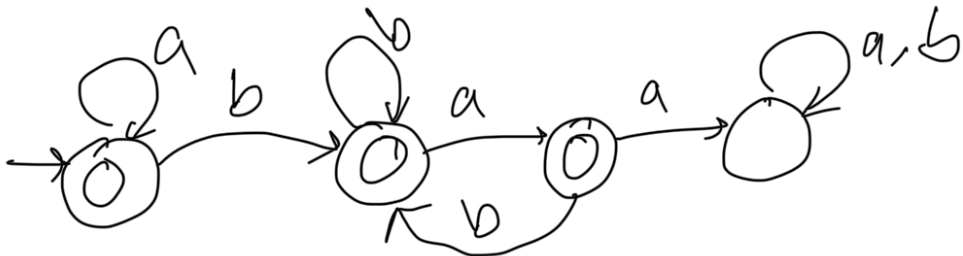
a)



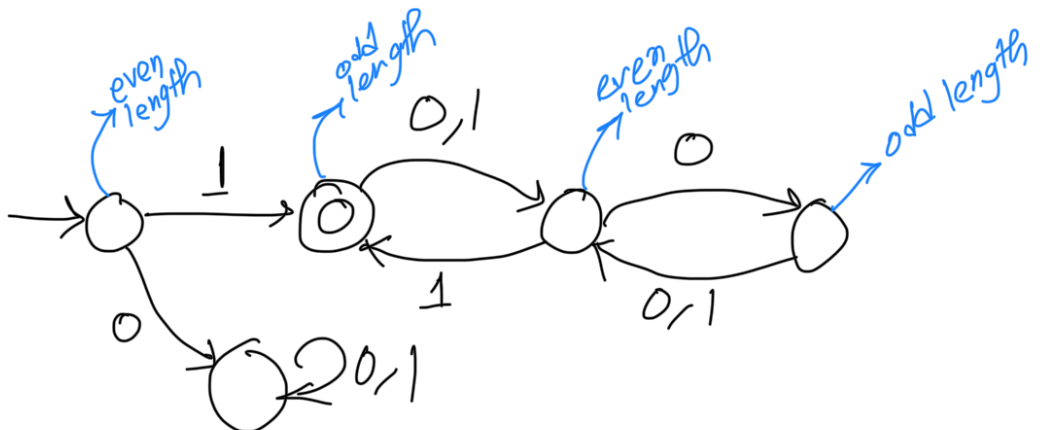
b)



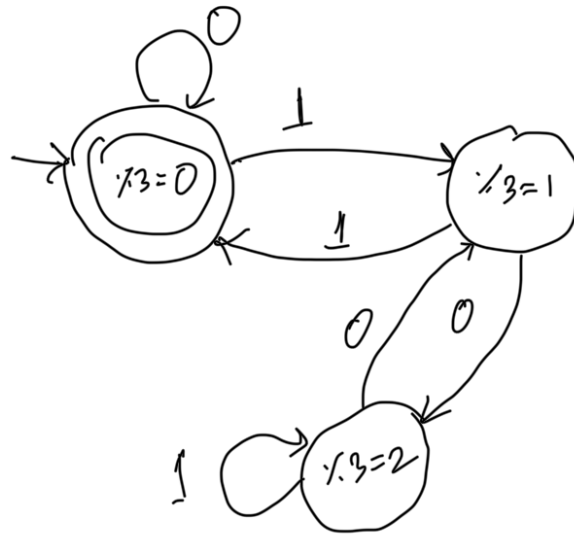
c)



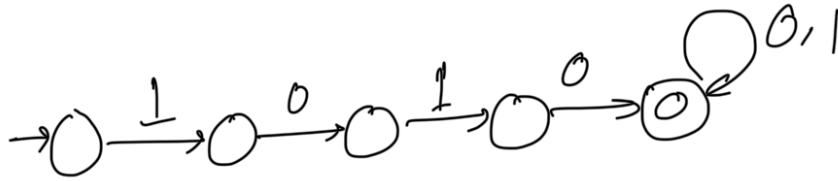
d)



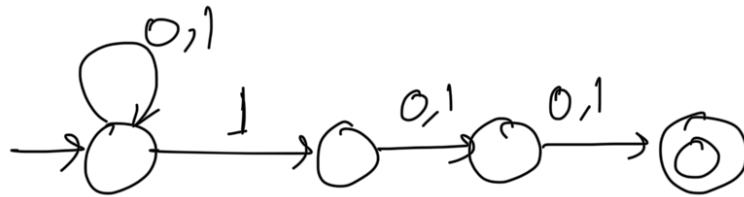
e)



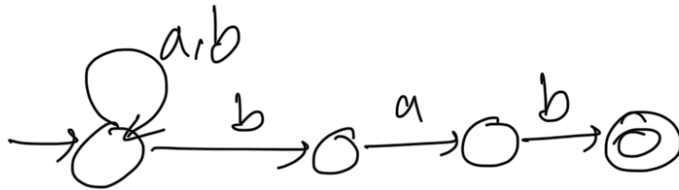
2. a)



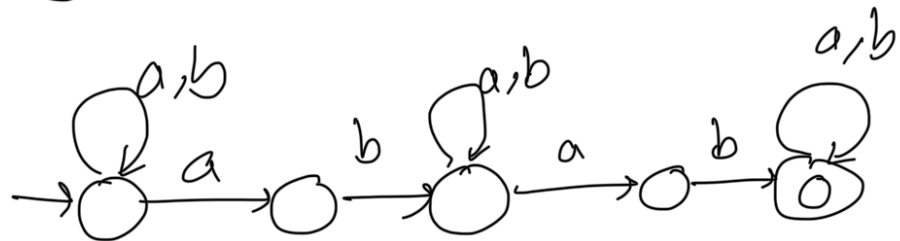
b)



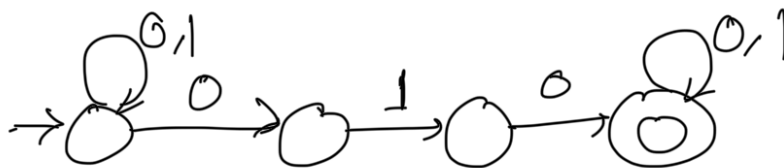
c)



d)



e)



3. a) $((0+1)(0+1))^*$

b) $a(a+b)((a+b)(a+b))^*a + b(a+b)((a+b)(a+b))^*b$
 $+ a + b$

c) $1^*0^*011^*0^*$

d) $a^*(baaa^*)^*$